

Instantaneous Center of Zero Velocity

INSTANTANEOUS AXIS OF ZERO VELOCITY

- The velocity of any point B located on the rigid body can be obtained directly by choosing the base point A to be a point that has zero velocity at the instant considered.
- For the velocity equation,

$$\mathbf{v}_B = \mathbf{v}_A + \boldsymbol{\omega} \times \mathbf{r}_{B/A}$$

$$\mathbf{v}_A = 0, \text{ the equation becomes,}$$

$$\mathbf{v}_B = \boldsymbol{\omega} \times \mathbf{r}_{B/A}$$

- For a body having general plane motion, point A so chosen is called the instantaneous center of zero velocity (IC), and it lies on the instantaneous axis of zero velocity.
 - This axis is always perpendicular to the plane of motion, and the intersection of the axis with this plane defines the location of the IC.
 - The equation becomes,

$$\mathbf{v}_B = \boldsymbol{\omega} \times \mathbf{r}_{B/IC}$$

and the point B moves momentarily about the IC in a circular path; rotation about the instantaneous axis.

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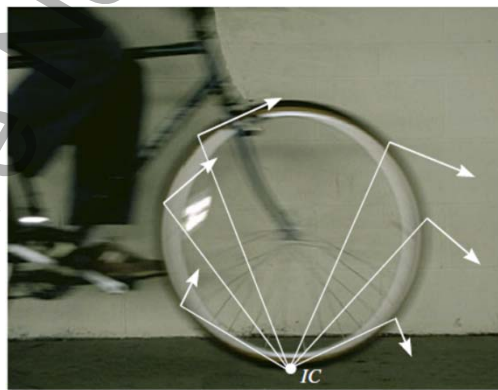
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- Example of a bicycle



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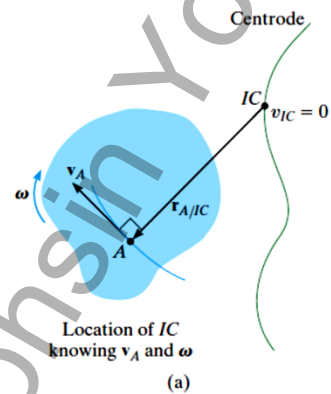
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Location of the IC. To locate the IC we can use the fact that the velocity of a point on the body is *always perpendicular* to the relative-position vector directed from the IC to the point. Several possibilities exist:

Case I:

The velocity $\mathbf{v}_A$ of a point A on the body and the angular velocity ω of the body are known, Fig. 16–18a. In this case, the IC is located along the line drawn perpendicular to $\mathbf{v}_A$ at A , such that the distance from A to the IC is $r_{A/IC} = v_A/\omega$. Note that the IC lies up and to the right of A since $\mathbf{v}_A$ must cause a clockwise angular velocity ω about the IC.



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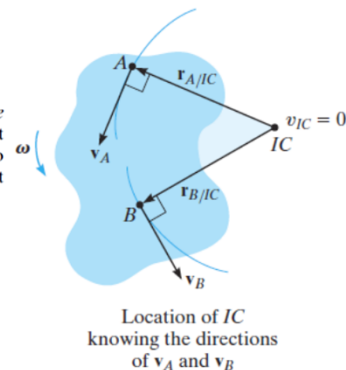
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Location of the IC. To locate the IC we can use the fact that the velocity of a point on the body is *always perpendicular* to the relative-position vector directed from the IC to the point. Several possibilities exist:

Case II:

The lines of action of two nonparallel velocities $\mathbf{v}_A$ and $\mathbf{v}_B$ are known, Fig. 16–18b. Construct at points A and B line segments that are perpendicular to $\mathbf{v}_A$ and $\mathbf{v}_B$. Extending these perpendiculars to their point of intersection as shown locates the IC at the instant considered.



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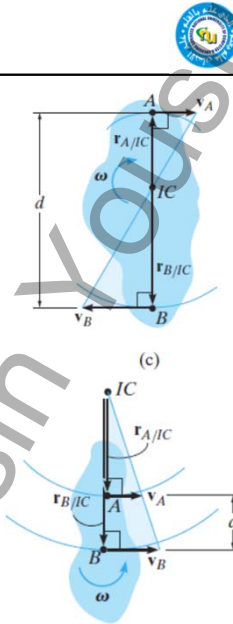
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INSTANTANEOUS AXIS OF ZERO VELOCITY

Location of the IC. To locate the IC we can use the fact that the *velocity* of a point on the body is *always perpendicular* to the *relative-position vector* directed from the IC to the point. Several possibilities exist:

Case III:

The magnitude and direction of two parallel velocities $\mathbf{v}_A$ and $\mathbf{v}_B$ are known. Here the location of the IC is determined by proportional triangles. Examples are shown in Fig. 16–18c and d. In both cases $r_{A/IC} = v_A/\omega$ and $r_{B/IC} = v_B/\omega$. If d is a known distance between points A and B , then in Fig. 16–18c, $r_{A/IC} + r_{B/IC} = d$ and in Fig. 16–18d, $r_{B/IC} - r_{A/IC} = d$.



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PROCEDURE FOR ANALYSIS

The velocity of a point on a body which is subjected to general plane motion can be determined with reference to its instantaneous center of zero velocity provided the location of the IC is first established using one of the three methods described above.

- As shown on the kinematic diagram in Fig. 16–19, the body is imagined as “extended and pinned” at the IC so that, at the instant considered, it rotates about this pin with its angular velocity ω .
- The *magnitude* of velocity for each of the arbitrary points A , B , and C on the body can be determined by using the equation $v = \omega r$, where r is the radial distance from the IC to each point.
- The *line of action* of each velocity vector $\mathbf{v}$ is *perpendicular* to its associated radial line $\mathbf{r}$, and the velocity has a *sense of direction* which tends to move the point in a manner consistent with the angular rotation ω of the radial line, Fig. 16–19.

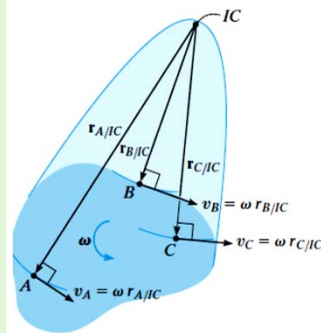


Fig. 16–19

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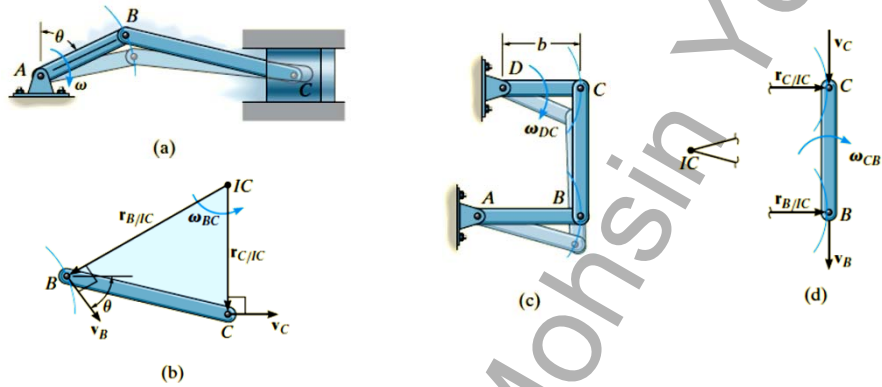
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EXAMPLES

EXAMPLE 16.10

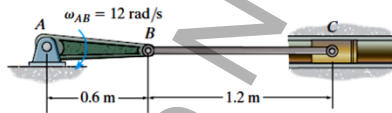
Show how to determine the location of the instantaneous center of zero velocity for (a) member *BC* shown in Fig. 16–20*a*; and (b) the link *CB* shown in Fig. 16–20*c*.



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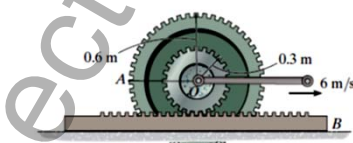
FUNDAMENTAL PROBLEMS

F16–14. Determine the angular velocity of link *BC* and the velocity of the piston *C* at the instant shown.



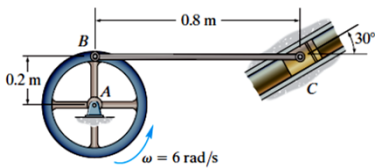
F16–14

F16–15. If the center *O* of the wheel is moving with a speed of $v_O = 6 \text{ m/s}$, determine the velocity of point *A* on the wheel. The gear rack *B* is fixed.

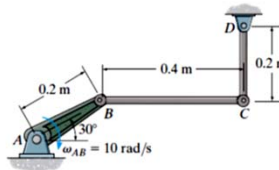


F16–15

F16–17. Determine the angular velocity of link *BC* and the velocity of the piston *C* at the instant shown.



F16–18. Determine the angular velocity of links *BC* and *CD* at the instant shown.

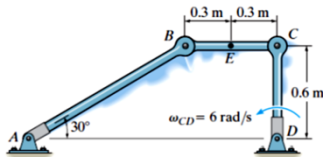


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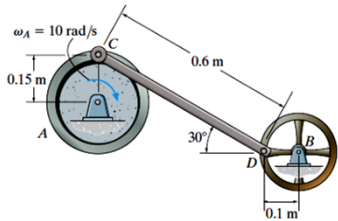


PRACTICE PROBLEMS

•16-89. If link CD has an angular velocity of $\omega_{CD} = 6 \text{ rad/s}$, determine the velocity of point E on link BC and the angular velocity of link AB at the instant shown.

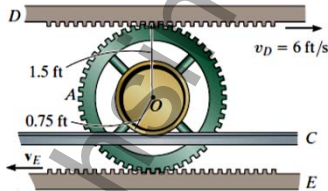


*16-104. If flywheel A is rotating with an angular velocity of $\omega_A = 10 \text{ rad/s}$, determine the angular velocity of wheel B at the instant shown.



16-94. The wheel is rigidly attached to gear A , which is in mesh with gear racks D and E . If D has a velocity of $v_D = 6 \text{ ft/s}$ to the right and wheel rolls on track C without slipping, determine the velocity of gear rack E .

16-95. The wheel is rigidly attached to gear A , which is in mesh with gear racks D and E . If the racks have a velocity of $v_D = 6 \text{ ft/s}$ and $v_E = 10 \text{ ft/s}$, show that it is necessary for the wheel to slip on the fixed track C . Also find the angular velocity of the gear and the velocity of its center O .



Probs. 16-94/95